Proteus Automation System

Network Proteus UI Developers Guide

Date

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Document Revision History

The revision history shows the history for this Product Design Document and provides descriptions of changes made.

Date	Version	Section/Page	Update Description	Contact Name

Contributors

The content of this document has been authored with the combined input of the following group of key individuals.

Name	Role	Area

Document Approval History

The section tracks the review and approval of this Product Design Document.

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Overview of this document

Network Proteus UI document is a quick onboarding guide for Angular developers. It provides step-by-step instructions to set up the developer's environment and code samples to develop and run the application. This document includes code samples, code flow diagrams and project related document links for clarity.

About Network Proteus

Network Proteus application automates various tasks in Load Balancing, Networking, and DNS administration that are manually handled by system engineers/administrators with minimal levels of automation. The purpose of Network Proteus is to:

- Create, update and close RFC tickets by executing GitLab CI/CD pipeline
- Configure various systems as per Playbooks created in AWX
- Create various interfaces to integrate systems. Example., Microservice to identify the status of Job in AWX and update in ServiceNow through GitLab CI/CD
- Rollback to original configuration in case of failure

Modules

Network Proteus has two modules:

- Load Balancer - You can manage Pool Member Information (Add, Modify and Delete)
- Networking- You can upload ACL Config Files and Schedule Jobs

*This document explains Code samples of Networking module.

Project References

The technical documentation stack is available on SharePoint. It includes these documents:

- Requirements Document
- Current State vs End State Assessment
- Design Document
- Dependencies Mapping
- Use Cases
- Test Plan and Results
- Instructional Manual and User Manual

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Tools

Gitlab CI/CD, AWX, Hashicorp vault, PKS (Kubernetes container) are the DevOps tools that are used in the Network Proteus application. You can follow the instructions below to setup the environment:

Gitlab

Setting up the environment

You must log into https://gitlab.dell.com and request access to the infrastructure/network-proteus/r-d-projects/networkproteus-ui group. The individual feature repositories are SN001, LB001 and NW001.

Prepare Your Development Environment

To build and run Network Proteus, you must install these software tools on your Windows development machine:

1. Download and install Microsoft Teams to collaborate with team members.
2. Download and install Git to maintain your own GitLab instance.
3. Request access to GitLab repository for Network-Proteus.
4. Configure SSH for GitLab. You can use the installed Git Bash.
5. To launch git bash, right click on the desktop and click git bash. See GitLab and SSH keys.

Download Node.js

To install Node.js, you need elevated access.

1. Raise an IT connect Ticket. For a long-term local administrator rights (One Year), submit the Local Admin Access [Auto] form.
2. Once it is approved, open the software center and install Local Admin access.

Install Node.js

Once your request is approved, Install Node.js on your local machine or computer.

Once installed, we can check it using the below command:

```
node –version
```

Once Node.js is installed npm is also installed. If you want to verify it type the below command:

```
npm -v
```

Install Angular cli using npm

```
npm install -g @angular/cli@^8.0.0
```


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Development Tools, Related Software and Hardware

The development tools used for Network Proteus include hardware, operating systems, and tools for program development, source control, general design, database design, debugging, defect tracking, and testing:

Infrastructure	UI	Backend	Testing	Devops	Testbed
Dell Digital Cloud PKS	Angular Node.js	Python 3x	pytest	Ansible	F5 Devices
Gitlab (Code Management)	Html/CSS	Flask	postman	Gitlab CI/CD	IOS Devices / Images
Docker	Bootstrap Ngx-bootstrap	MongoDB	Dell Customized tools	Hashi Corp Vault	DELL OS 6
Service Now (CMDB)			SaaS Labs Virtual environment		DELL OS 9
Dell Digital Cloud VMs			Tox		DELL 10 Virtual
Dell Digital Cloud Object Storage (S3)			Coverage		ESXi Hosts and Storage
			SonarQube		Bluecat Testbed - Virtual
			Blackduck		Infoblox Testbed
			Checkmarx		

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Developer Guide and Code Sample

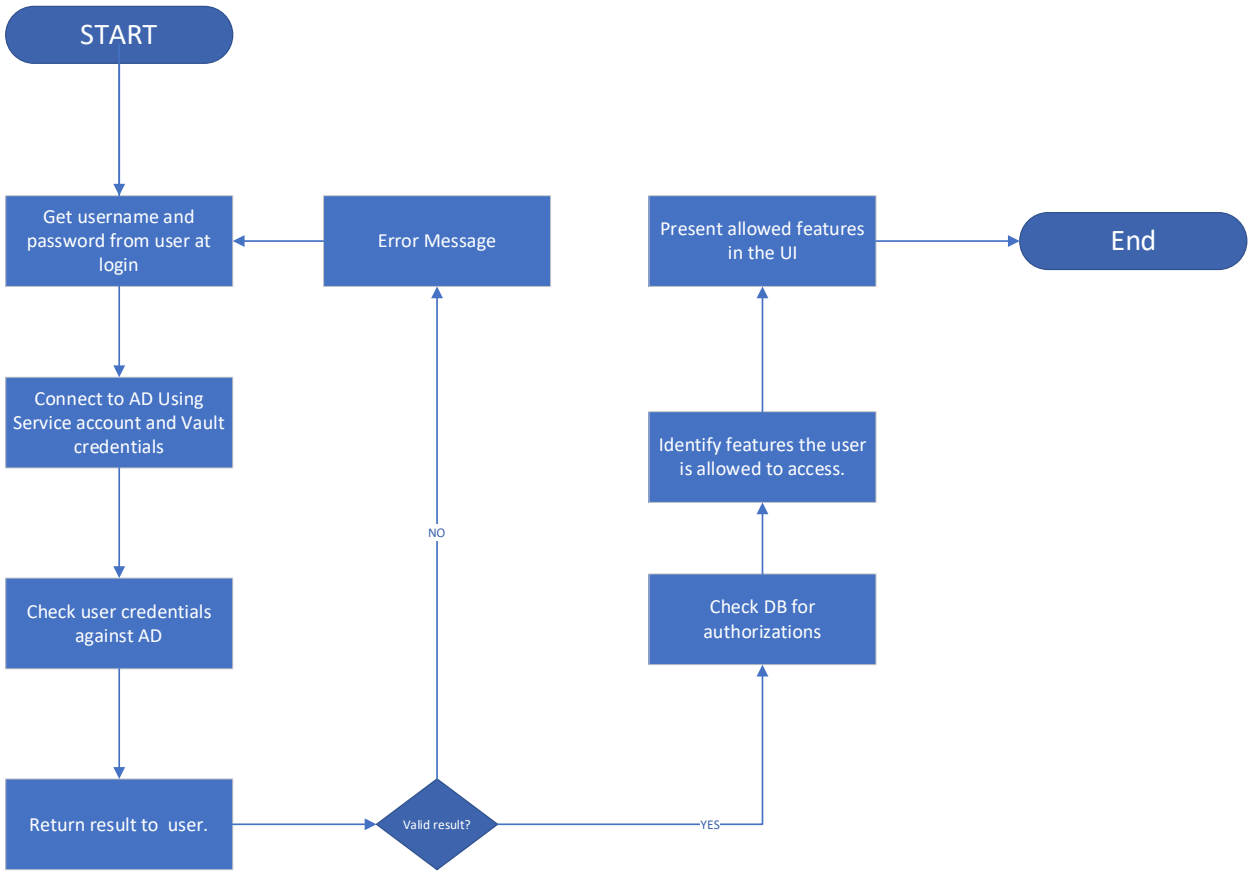
After setting up your development environment, you can reuse the Code samples to build and run Network Proteus UI.

Validate the User Credentials

Once you enter the Dell credentials on the login screen, the system authorizes the user against the Active Directory (AD), using the user ID and password.

Code Flow

This Code flow explains the user authorization function.



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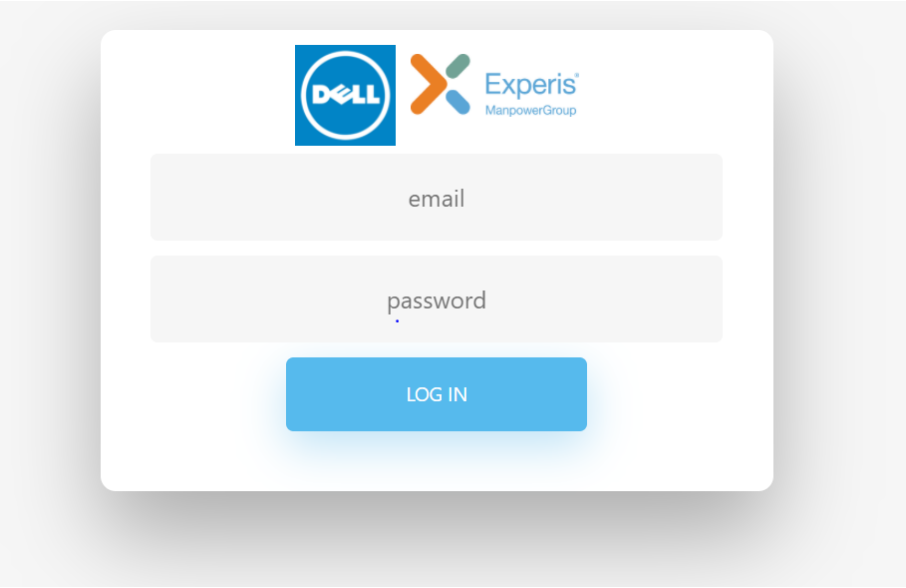
Get User Role



Use submitLogin(email, password) to check the user credentials against AD role and identify the features the user is privileged to access. For this we use loginAPI.

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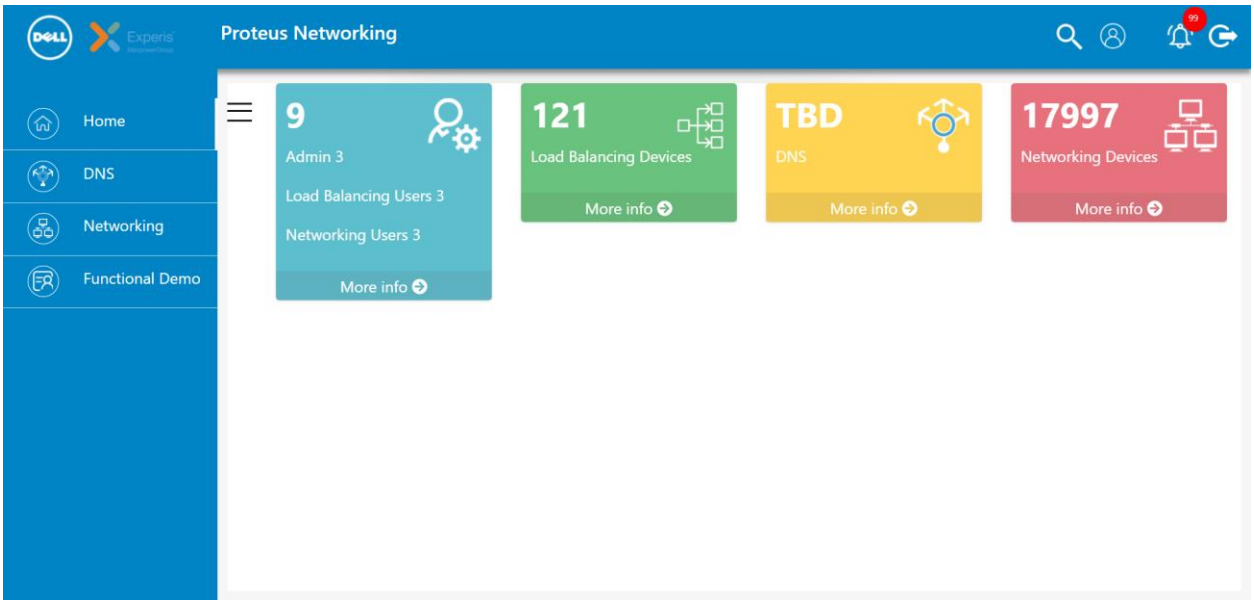
Dashboard



After the successful login, the system displays the following information:

- Total Number of users from Mongo DB
- Total Number of load balancing devices from Mongo DB
- Total Number of network devices from Mongo DB

Header Component

Used to display all the necessary assets in the header (for e.g, dell and experis logo, network proteus title, search bar, current user and logout.)

1) **Search option**:- Search option to search by request ID.
2) **Current user:-** Popup the application user name and User role.
3) **Logout:**- Logout from the application and redirect to login screen.

Header Component code:-

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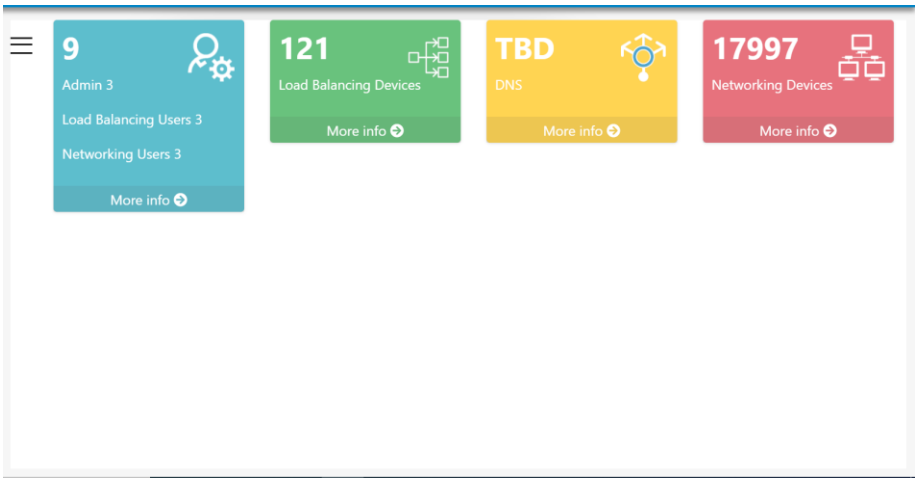
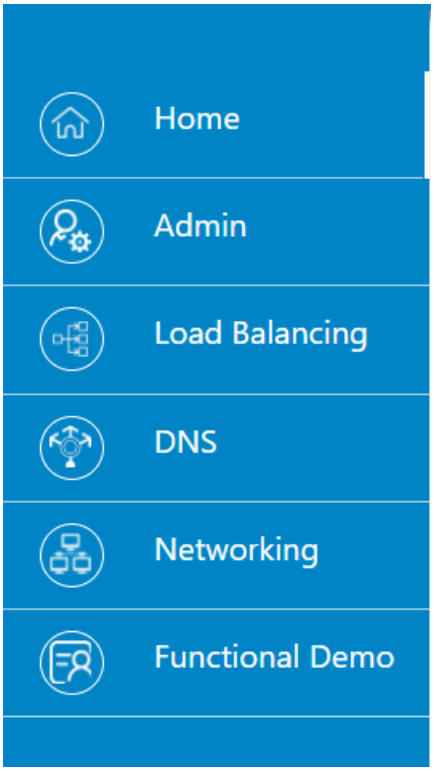
Home Component



Left Menu Component



After the user's successful login it will display the menu bar to the user's scope. Data services are used.

Application component show the access for the application according to user role.

Admin has the permission to access all the module of the application.

Load Balancing Users has access for only Home, Load Balancing and Functional demo.

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Networking Users has access for only Home, Networking and Functional demo.

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Admin Component Table with search : -

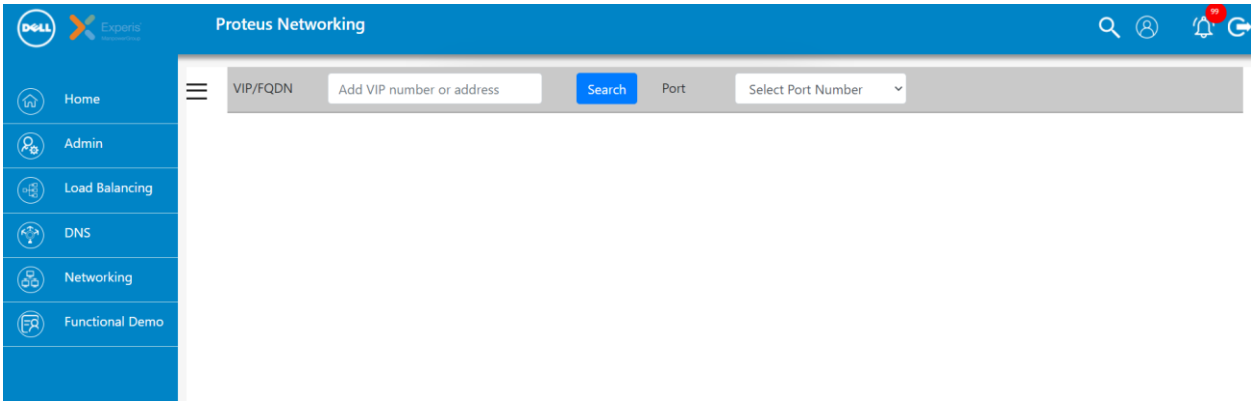
Job Status | User Request

Full text search:

#	Request ID	Feature ID	Pipeline ID	Requested By	Start Date
	2111115595	LB001	7206431	sarath.chintapatla@dellteam.com	11/11/21, 8:25 PM
	2111115526	SN001	7205919	sarath.chintapatla@dellteam.com	11/11/21, 7:50 PM
	21110864015	SN001	7154368	ashwini.kalyanpur@dellteam.com	11/8/21, 6:38 PM
	21110830921	SN001	7152867	annam.kumar@dellteam.com	11/8/21, 5:07 PM
	21110879630	SN001	7152363	annam.kumar@dellteam.com	11/8/21, 4:35 PM
	21110768665	LB001	7143293	annam.kumar@dellteam.com	11/7/21, 12:21 PM
	2111074802	SN001	7143288	annam.kumar@dellteam.com	11/7/21, 12:16 PM
	21110772033	SN001	7143273	annam.kumar@dellteam.com	11/7/21, 12:06 PM
	2111054014	SN001	7135160	sarath.chintapatla@dellteam.com	11/5/21, 9:37 PM

Load Balancing Component:-

Search FQDN or IP Address:-



Onclick of serach button providing the FQDN, select poart and Select pool members. Will show the screen related to the perticular FQDN with the option of Show last 10 result, Add Poolmember, Edit pool member and Delete pool member.

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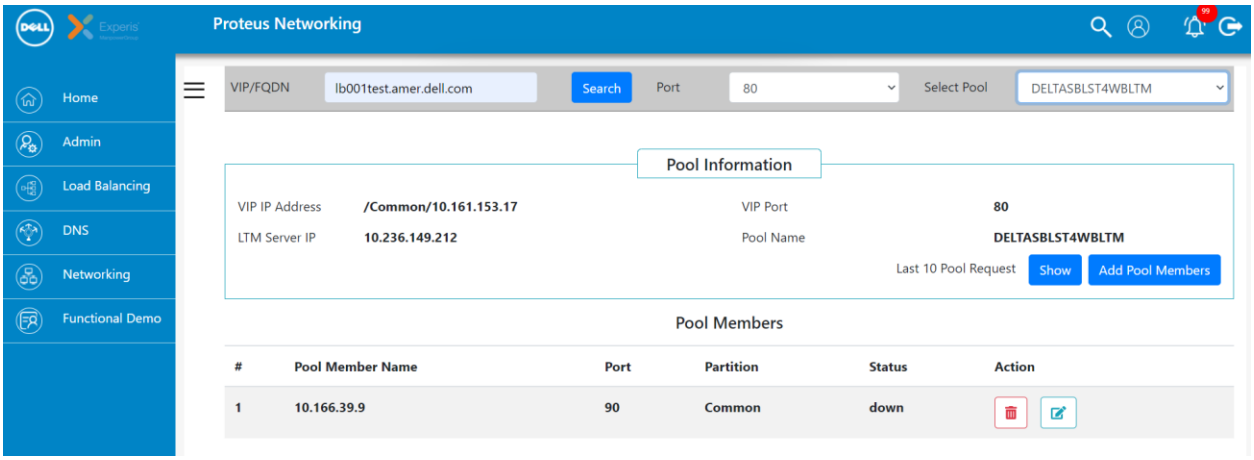
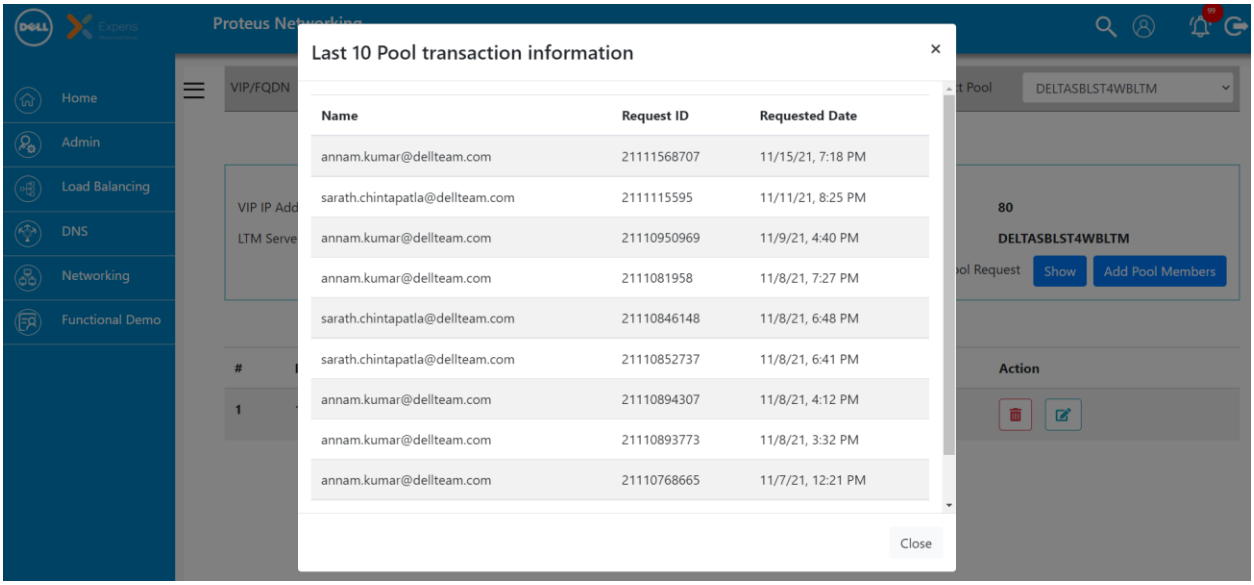


Show last 10 result:-



, Add Pool member:-

Has option to add new pool member with IP address configerable

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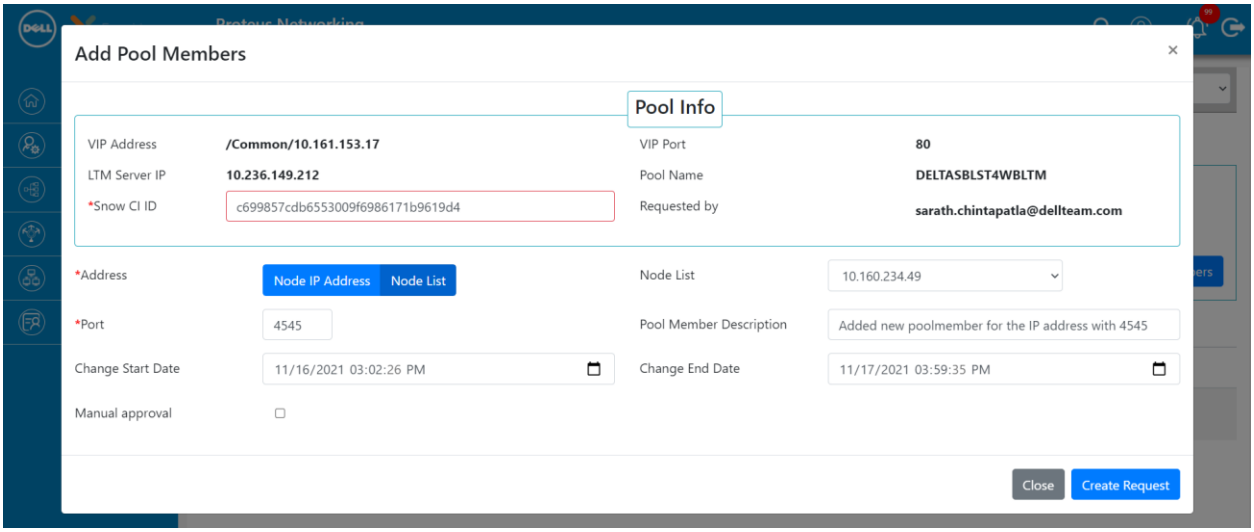
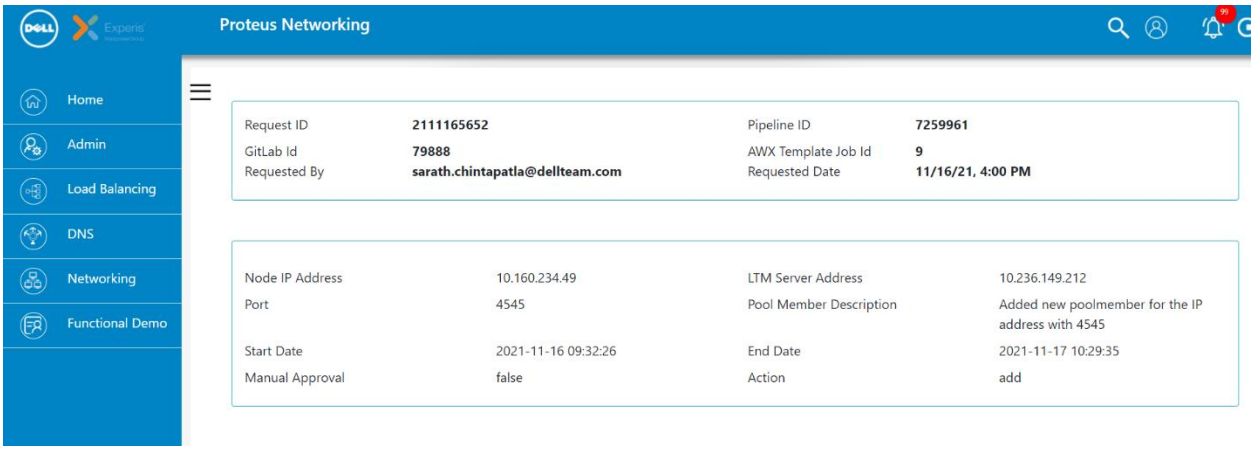


Creating new add request for the IP address:-



Which runs the Git lab CICD pipeline process.

CI\CD AWX SNOW

#	GitLab Job ID	GitLab Pipeline Id	Status	Stage
1	55810980	7259961	success	rfc-submit
2	55810981	7259961	running	ansible-playbook
3	55810982	7259961	created	rfc-update
4	55810983	7259961	created	rfc-close
5	55810984	7259961	created	rfc-check

AWX job id created successful:-

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CI\CD AWX SNOW

#	Schedule Id	Job ID	Job Status
1	940	3069	successful

Service now ticket generated:-

CI\CD AWX SNOW

#	CR No	CR Status
1	"CHG0410337"	Started

On successful of pipeline completed it has the option of roll back or close and also you can verify the Original and changed config file :-

CI\CD AWX SNOW

#	CR No	CR Status
1	"CHG0410337"	Started

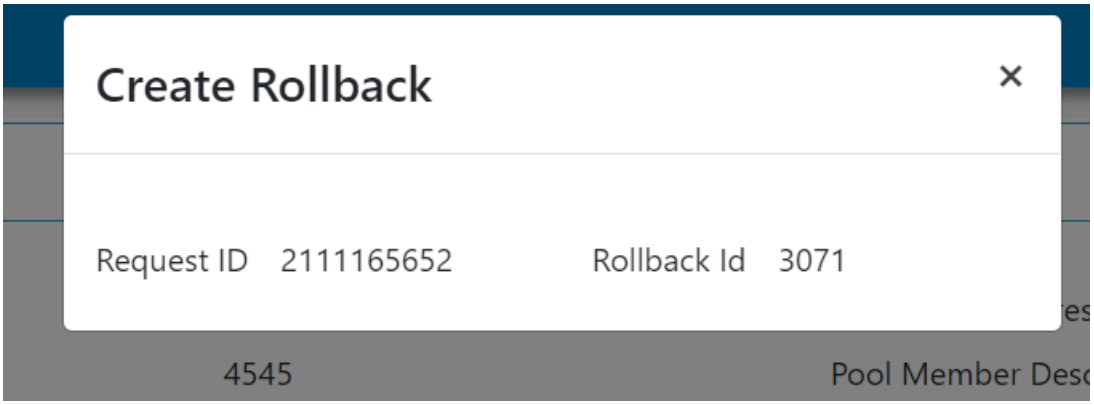
RFC Rollback or Close

Rollback Close RFC

Configuration Status

Original Config Changed Config

Roll back click create one more AWX rollback request ID:-



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CI\CD | AWX | SNOW

#	Schedule Id	Job ID	Job Status
1	940	3069	successful
1	940	3071	successful

Original config click:-

Standard Changes

```
}
/Common/10.166.44.15:1636 {
address 10.166.44.15
}
}
monitor /Common/Tcp_1389
}
ltm pool /Common/oisperf_80 {
members {
/Common/10.160.234.49:80 {
address 10.160.234.49
}
/Common/10.160.234.50:80 {
address 10.160.234.50
}
}
monitor /Common/http
}
ltm pool /Common/ombudsuat-bigip-80 {
```

Close

Changed config shows the actual tried to add the pool member in UI:-

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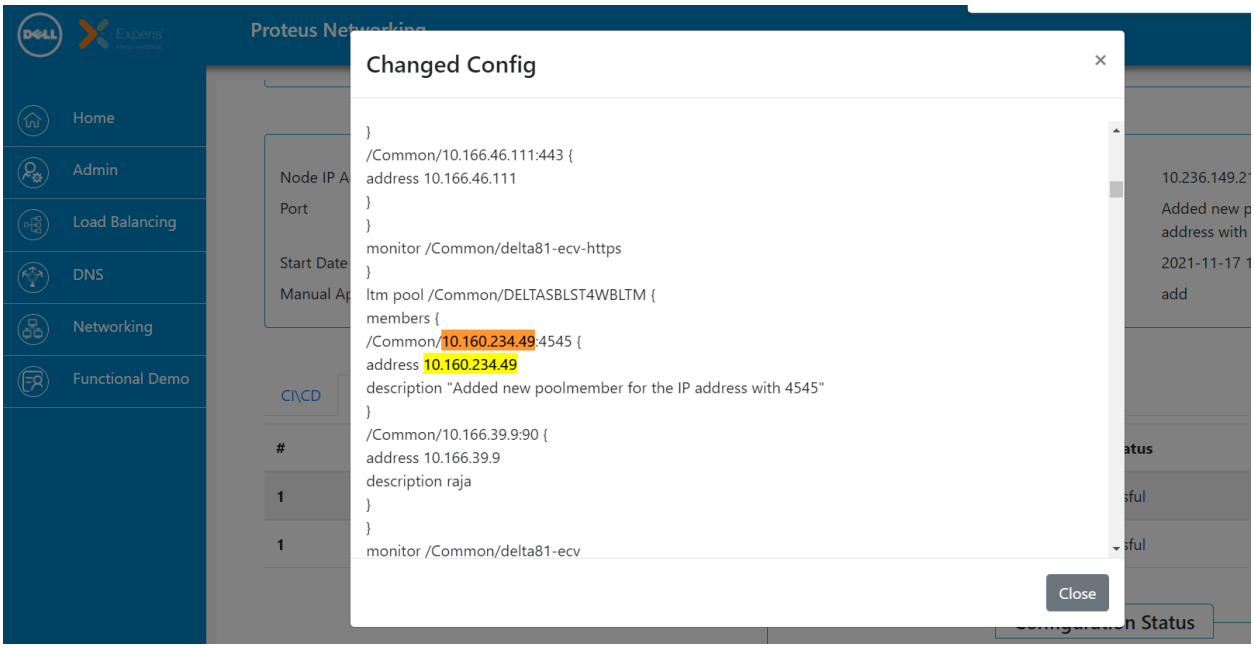
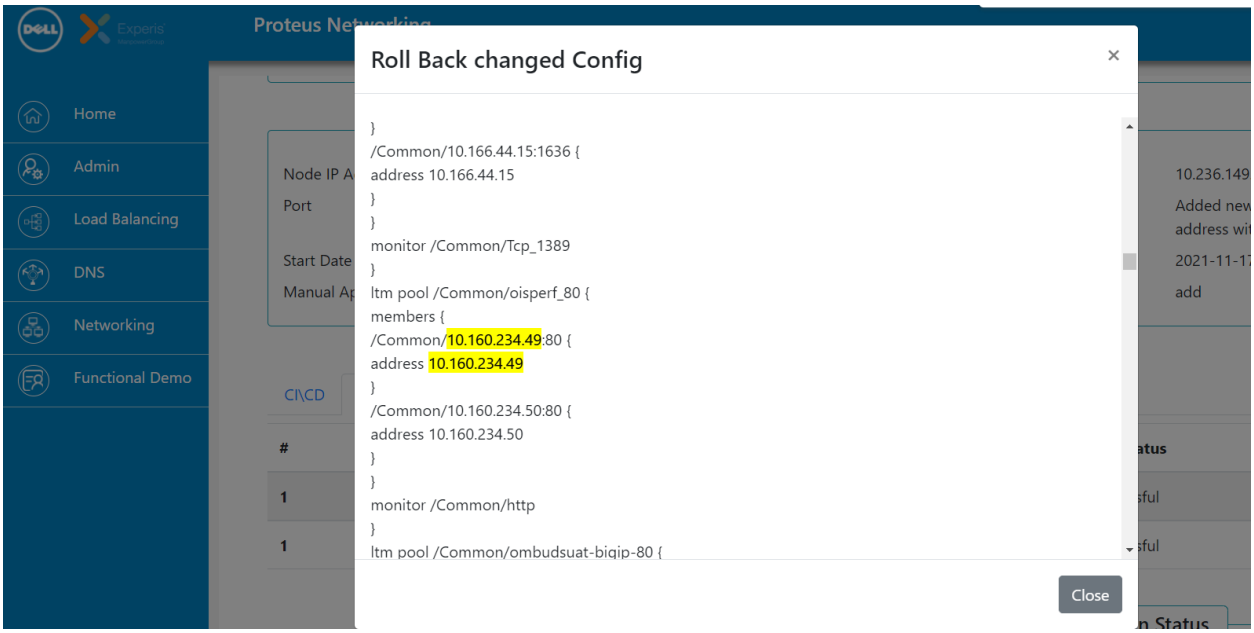


As we tried to roll back it shows the original file back will not add the pool in UI:-



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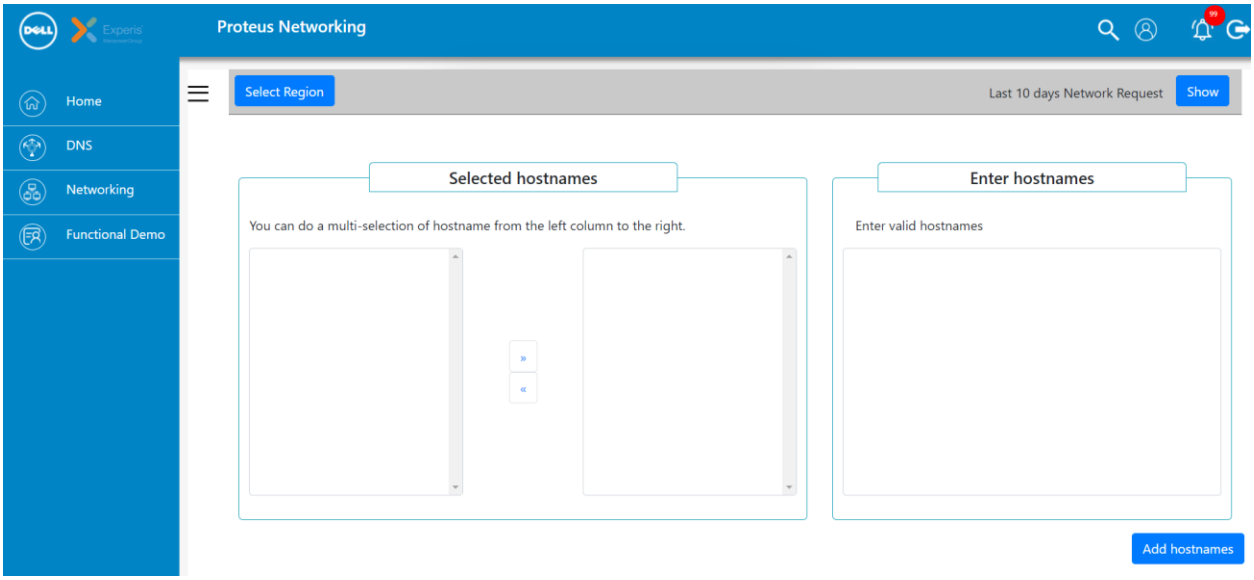
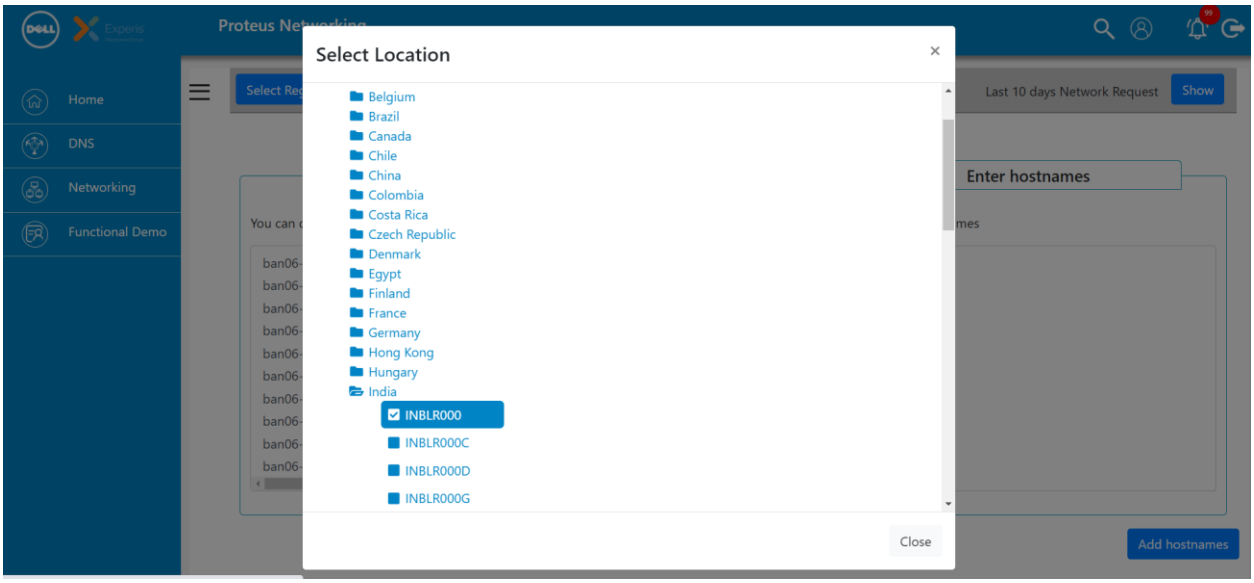
Networking Component :-



Need to select region which gives the different location for selecting the location to trigger the networking pipeline.



Once select the region device lest will be loaded in the box and select the device to configure.

As we have testing device provided by dell for testing which can be selected from the list(Tree node and device) INBLR17-DC-0118-OM02(OS 9) and USAUS01-MD-1501-SW02(OS 10) and for IOS we are adding rr2ecorpmcore1.

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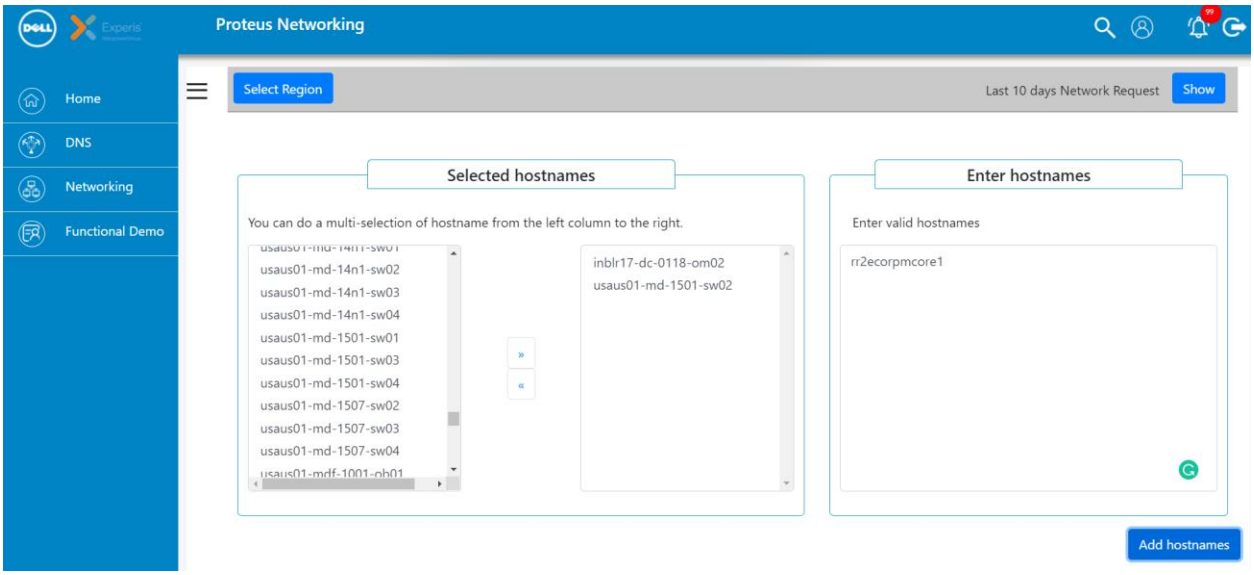
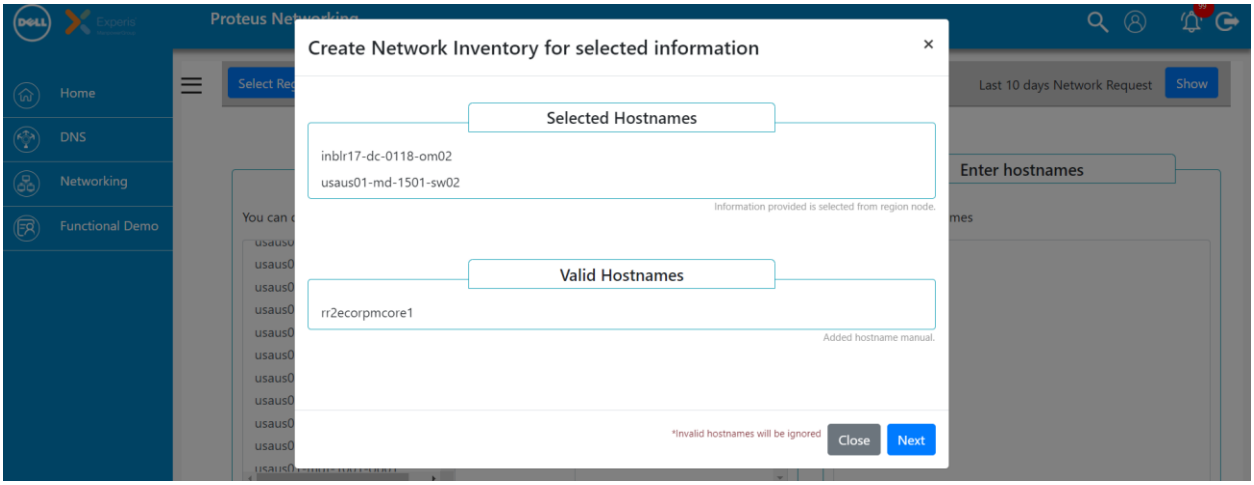


On click of Add hostname we can verify the device which we are trying to trigger



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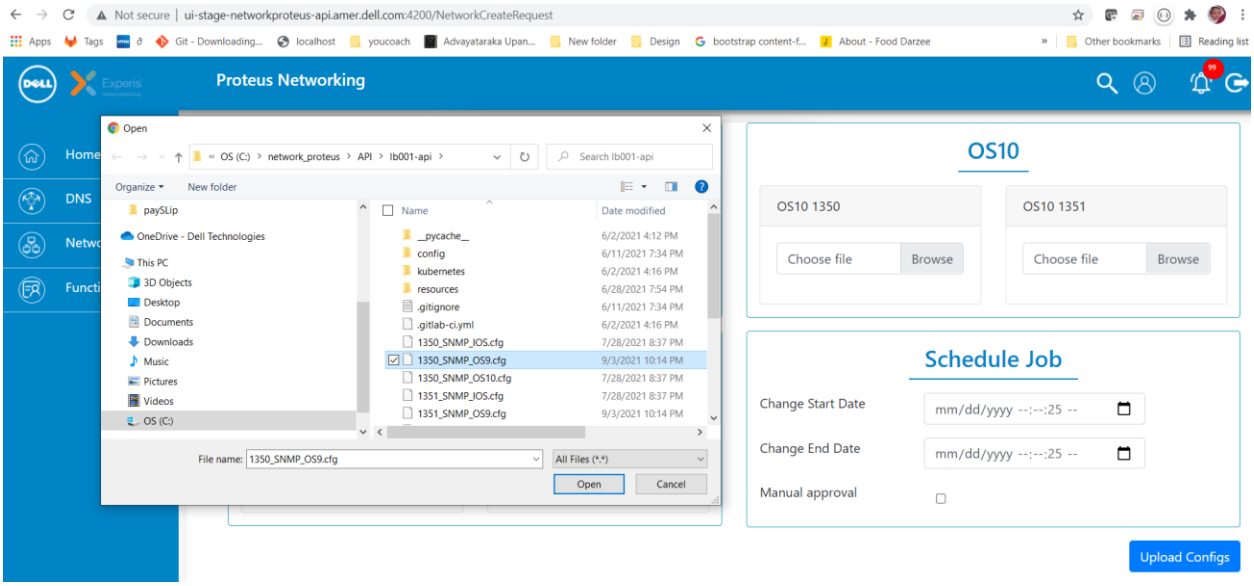
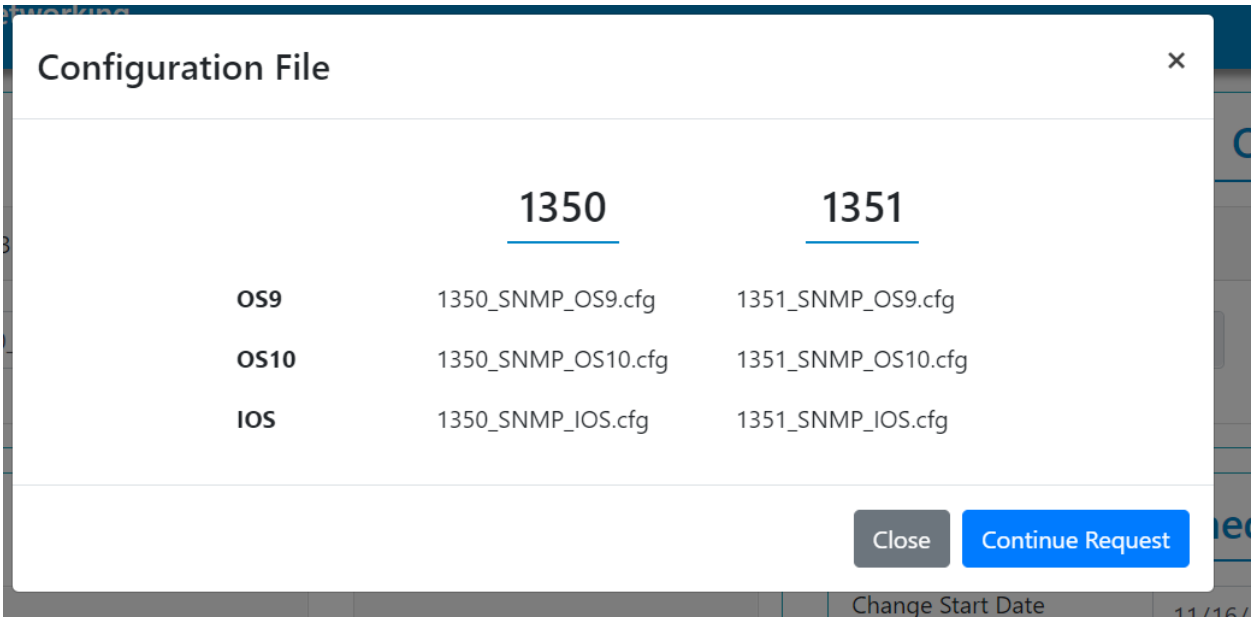
On click of next it allows to add config file to all three device (OS9, OS10 and IOS) .

Browse and upload files for all device.
Can add the date and time to schedule trigger the pipeline and end date and time also.

Also have option for manual approval to continue to run the pipeline.



Confirm the pipeline trigger.



Loads the pipeline trigger with CICD, AWX and SNOW.

It ask for manual trigger to continue on click of continue it starts the pipeline running.

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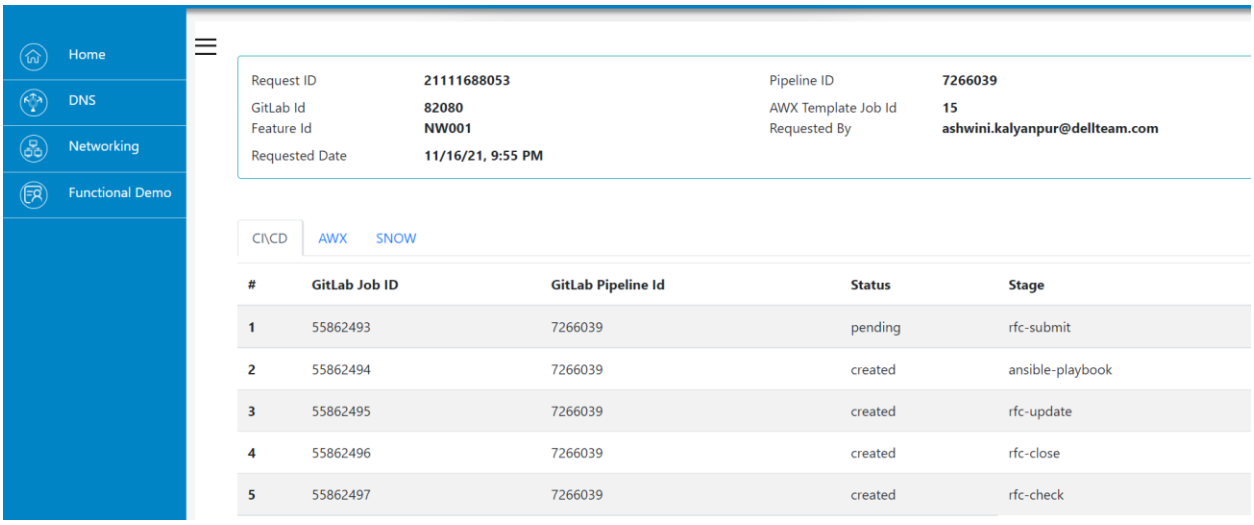
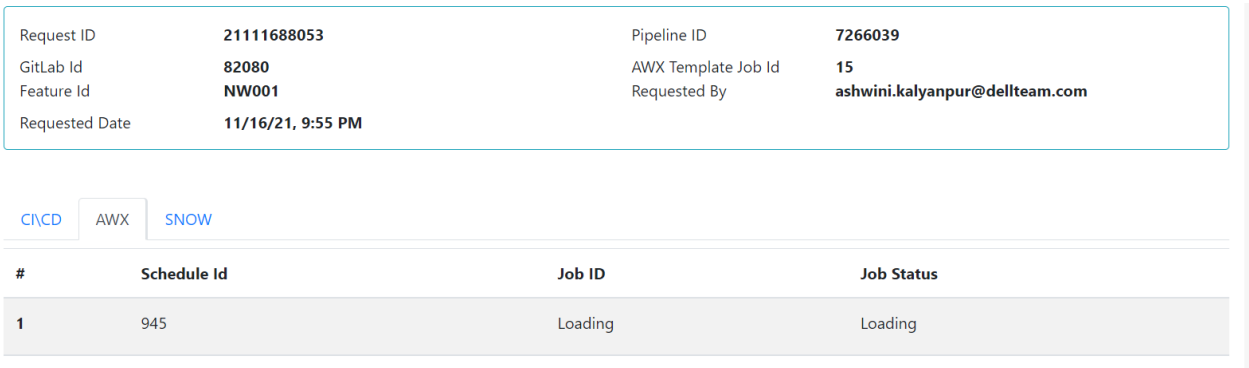
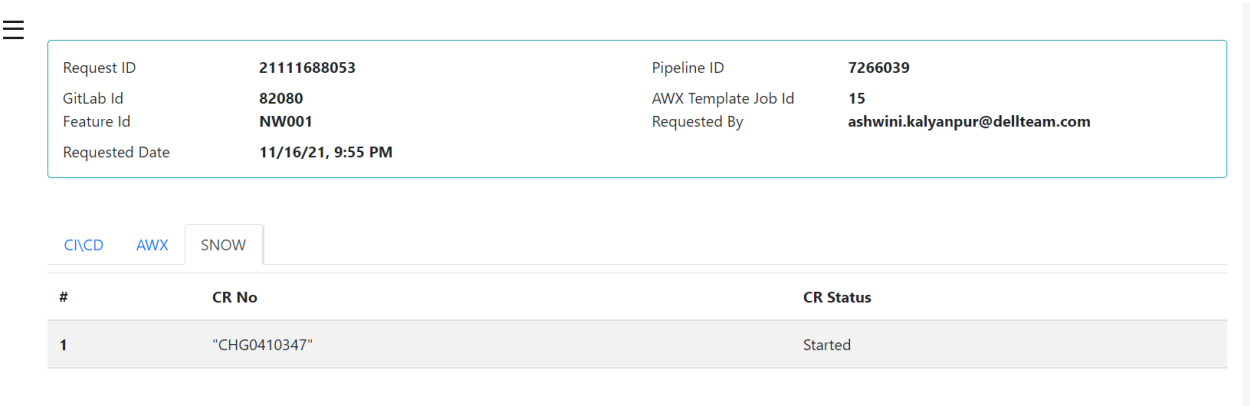


AWX schedule ID created



Created SNOW ticket



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Request ID	**21111688053**	Pipeline ID	**7266039**
GitLab Id	**82080**	AWX Template Job Id	**15**
Feature Id	**NW001**	Requested By	**ashwini.kalyanpur@dellteam.com**
Requested Date	**11/16/21, 9:55 PM**		

CI\CD AWX SNOW

#	GitLab Job ID	GitLab Pipeline Id	Status	Stage
1	55862493	7266039	success	rfc-submit
2	55862494	7266039	success	ansible-playbook
3	55862495	7266039	success	rfc-update
4	55862496	7266039	manual	rfc-close
5	55862497	7266039	created	rfc-check

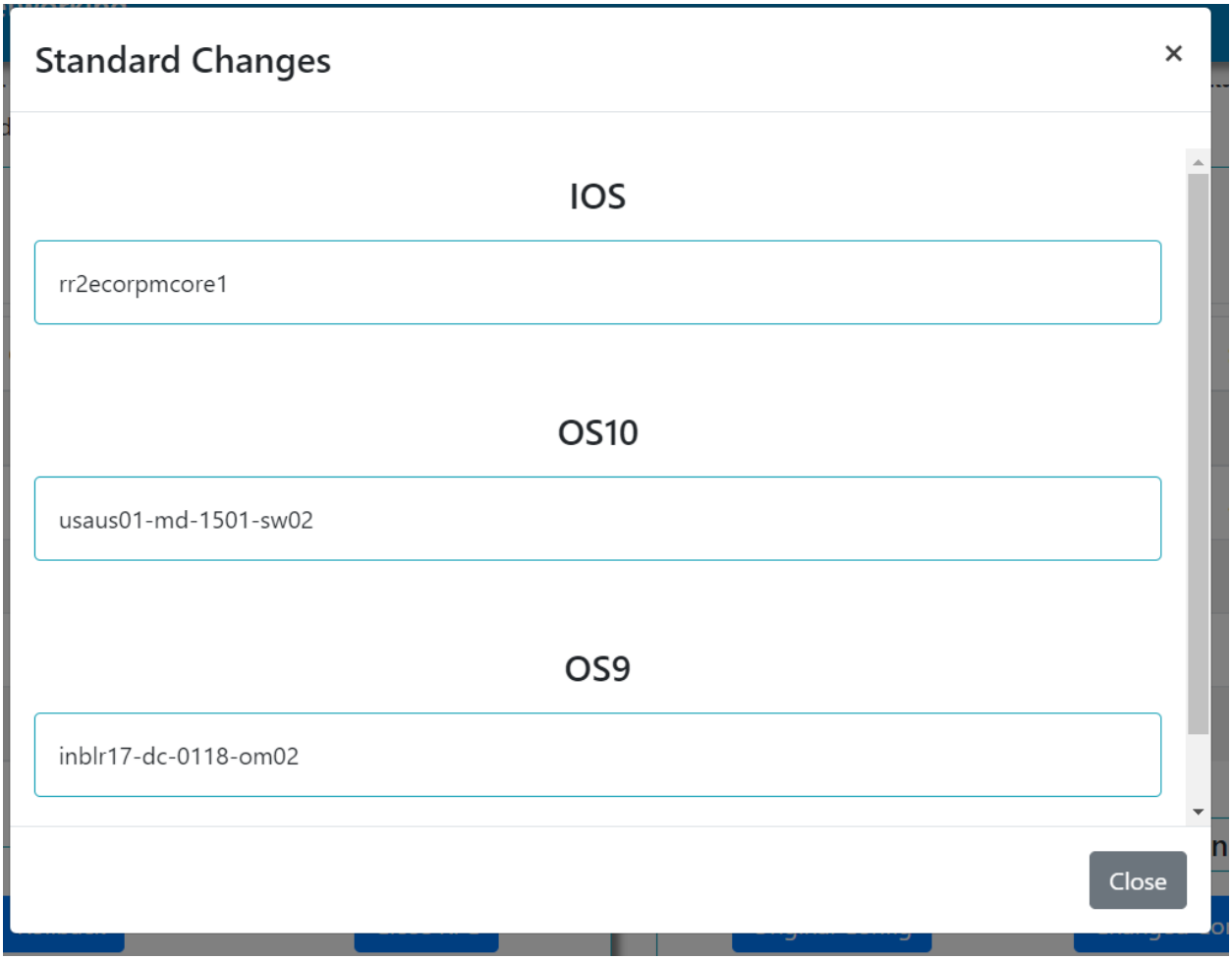
RFC Rollback or Close

Rollback Close RFC

Configuration Status

Original Config Changed Config

Shows the standard changes configure file for all three device.



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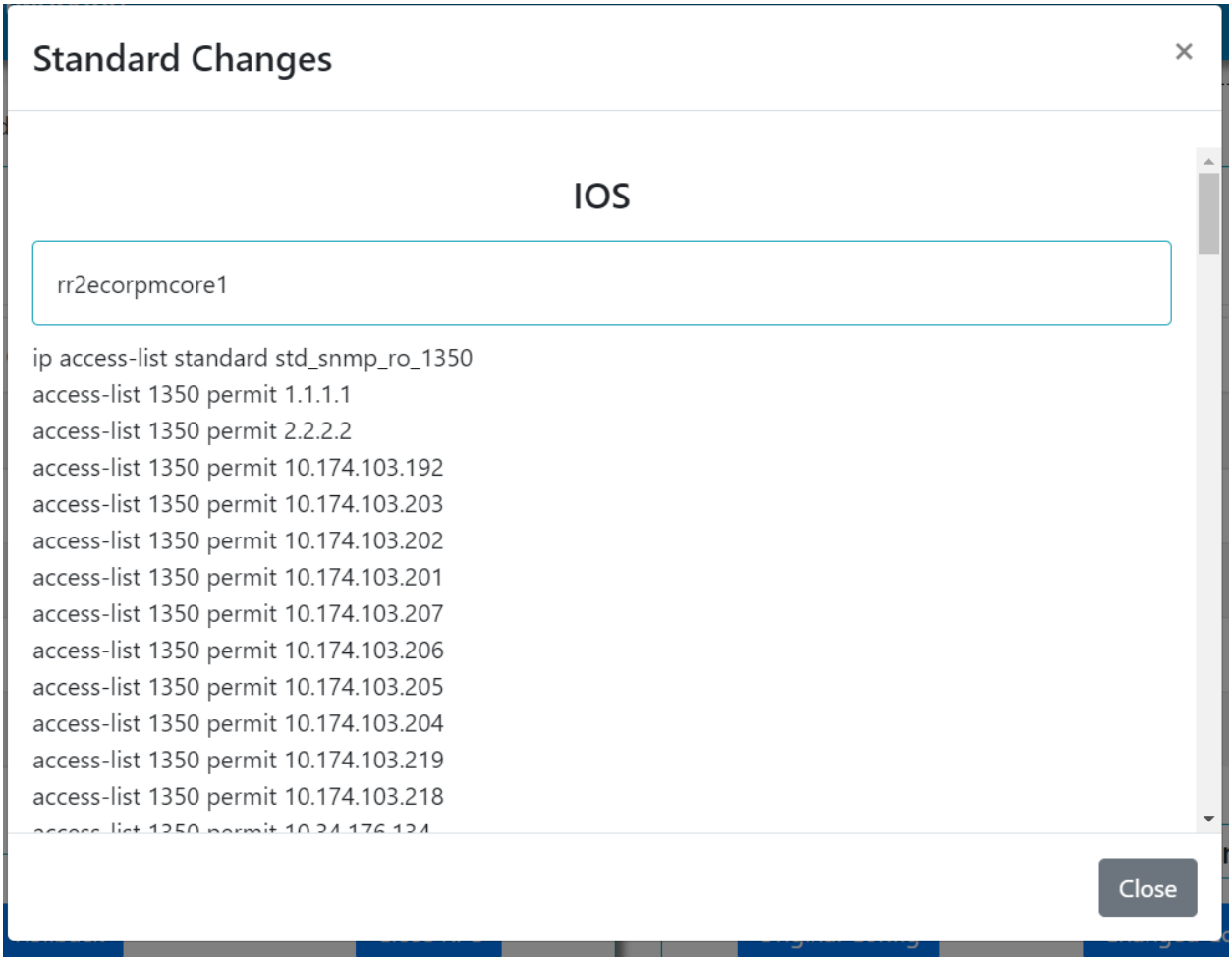

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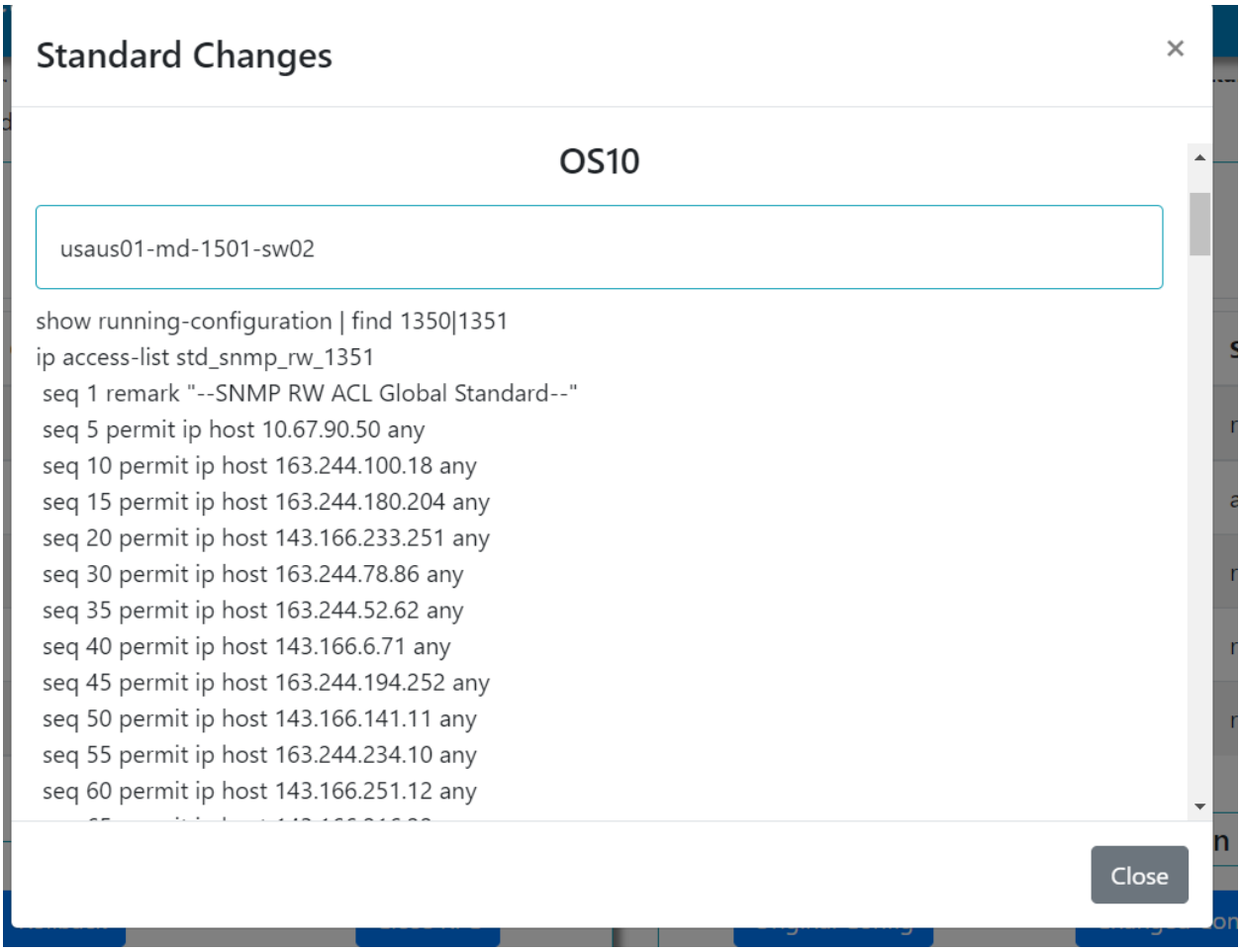

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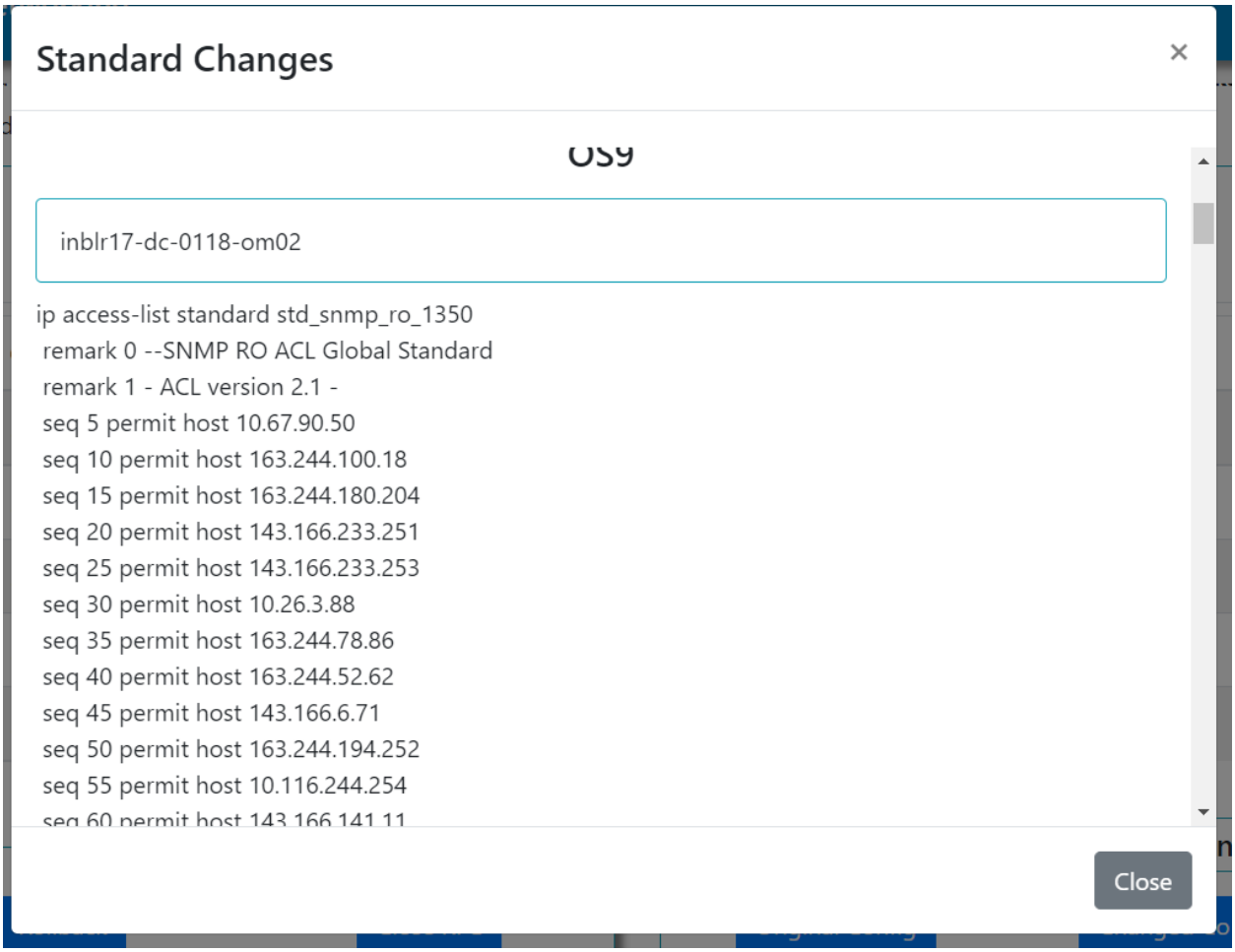

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